

Chapter 2

Descriptive Statistics

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

Ron Larson



Chapter Outline

- 2.1 Frequency Distributions and Their Graphs
- 2.2 More Graphs and Displays
- 2.3 Measures of Central Tendency
- 2.4 Measures of Variation
- 2.5 Measures of Position

Section 2.5

Measures of Position

Section 2.5 Objectives

- How to find the first, second, and third quartiles of a data set, how to find the interquartile range of a data set, and how to represent a data set graphically using a box-and whisker plot
- How to interpret other fractiles such as percentiles and how to find percentiles for a specific data entry
- Determine and interpret the standard score (z -score)

Quartiles

- **Fractiles** are numbers that partition (divide) an ordered data set into equal parts.
- **Quartiles** approximately divide an ordered data set into four equal parts.
 - **First quartile, Q_1** : About one quarter of the data fall on or below Q_1 .
 - **Second quartile, Q_2** : About one half of the data fall on or below Q_2 (median).
 - **Third quartile, Q_3** : About three quarters of the data fall on or below Q_3 .

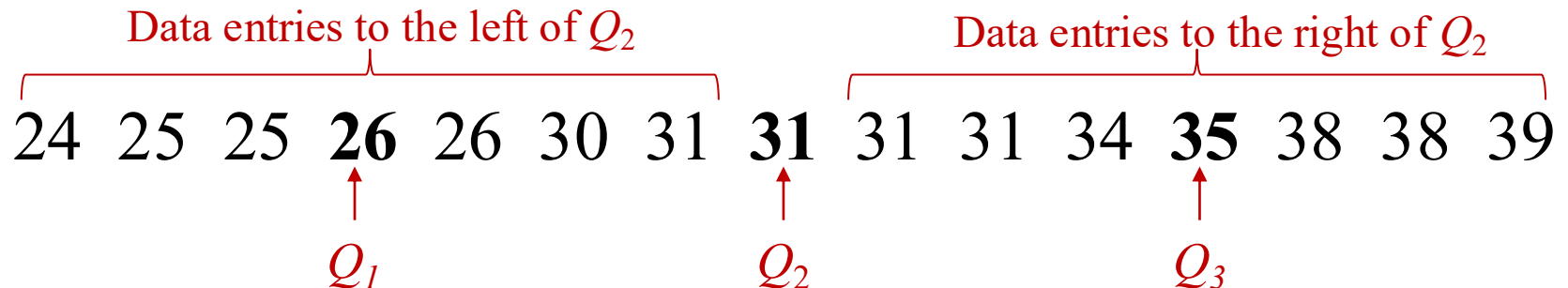
Example: Finding Quartiles

The amounts of fuel (in gallons per year) that automobile commuters waste due to traffic congestion in the 15 largest U.S. urban areas (where the populations are over 3 million) are listed. Find the first, second, and third quartiles of the data set. What do you observe? (*Source: Texas A&M Transportation Institute*)

34 30 31 31 25 31 25 24 38 26 39 26 38 31 35

Solution:

- Q_2 divides the data set into two halves.



Solution: Finding Quartiles

- In about one-quarter of the large urban areas, auto commuters waste 26 gallons of fuel or less, about one-half waste 31 gallons or less, and about three-quarters waste 35 gallons or less.

Example: Using Technology to Find Quartiles

The tuition costs (in thousands of dollars) for 25 liberal arts colleges are listed. Use technology to find the first, second, and third quartiles. What do you observe?

(Source: U.S. News & World Report)

55 59 55 56 57 55 60 59 61 58 57 61 59
48 44 30 39 58 48 46 19 55 45 48 48

Solution: Using Technology to Find Quartiles

Minitab and the TI-84 Plus each have features that calculate quartiles. Try using this technology to find the first, second, and third quartiles of the tuition data.

From the displays, you can see that

$$Q_1 = 47, Q_2 = 55, \text{ and } Q_3 = 58.5.$$

Solution: Using Technology to Find Quartiles

MINITAB

Descriptive Statistics: Tuition

Variable	N	Mean	SE Mean	StDev	Minimum
Tuition	25	51.20	2.04	10.18	19.00

Variable	Q1	Median	Q3	Maximum
Tuition	47.00	55.00	58.50	61.00

TI-84 PLUS

1-Var Stats

↑n=25
minX=19
Q₁=47
Med=55
Q₃=58.5
maxX=61

STATCRUNCH

Summary statistics:

Column	Q1	Median	Q3
Tuition	48	55	58

About one-quarter of these colleges charge tuition of \$47,000 or less; about one-half charge \$55,000 or less; and about three-quarters charge \$58,500 or less.

Interquartile Range

Interquartile Range (IQR)

- A measure of variation that gives the range of the middle portion (about half) of the data.
- The difference between the third and first quartiles.
- $IQR = Q_3 - Q_1$

Interquartile Range

Using the Interquartile Range to Identify Outliers

1. Find the first (Q_1) and third (Q_3) quartiles of the data set.
2. Find the interquartile range: $IQR = Q_3 - Q_1$.
3. Multiply IQR by 1.5: $1.5(IQR)$.
4. Subtract $1.5(IQR)$ from Q_1 . Any data entry less than $Q_1 - 1.5(IQR)$ is an outlier.
5. Add $1.5(IQR)$ to Q_3 . Any data entry greater than $Q_3 + 1.5(IQR)$ is an outlier.

Example: Finding the Interquartile Range

Find the interquartile range of the data set from the previous example. Are there any outliers?

Solution:

Recall $Q_1 = 47$ and $Q_3 = 58.5$. So, the interquartile range is $IQR = Q_3 - Q_1 = 58.5 - 47 = 11.5$.

To identify any outliers, first note that $1.5(IQR) = 1.5(11.5) = 17.25$.

Solution: Finding the Interquartile Range

- There is a data entry, 19, that is less than

$$Q_1 - 1.5(\text{IQR}) = 47 - 17.25 = 29.75$$

- A data entry less than 29.75 is an outlier.
- There are no data entries greater than

$$Q_3 + 1.5(\text{IQR}) = 58.5 + 17.25 = 75.75$$

- A data entry greater than 75.75 is an outlier.
- So, 19 is an outlier.

The costs of tuition and fees for the liberal arts colleges listed in the middle of the data set varies by at most \$17,250. Notice that the outlier, 19 (or \$19,000), does not affect the IQR.

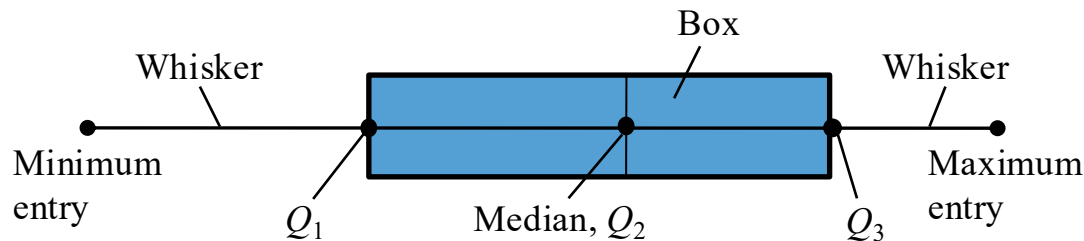
Box-and-Whisker Plot

Box-and-whisker plot

- Exploratory data analysis tool.
- Highlights important features of a data set.
- Requires (**five-number summary**):
 1. Minimum entry
 2. First quartile Q_1
 3. Median Q_2
 4. Third quartile Q_3
 5. Maximum entry

Drawing a Box-and-Whisker Plot

1. Find the five-number summary of the data set.
2. Construct a horizontal scale that spans the range of the data.
3. Plot the five numbers above the horizontal scale.
4. Draw a box above the horizontal scale from Q_1 to Q_3 and draw a vertical line in the box at Q_2 .
5. Draw whiskers from the box to the minimum and maximum entries.

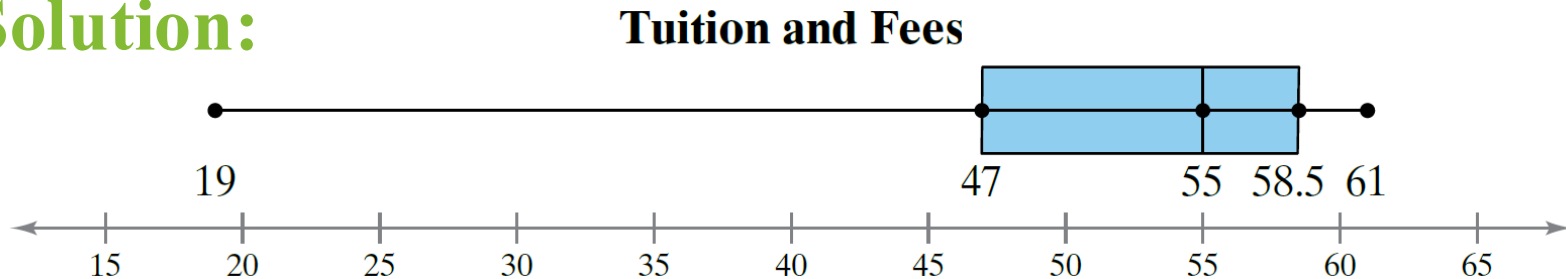


Example: Drawing a Box-and-Whisker Plot

Draw a box-and-whisker plot that represents the data set in the previous example.

Min = 19, $Q_1 = 47$, $Q_2 = 55$, $Q_3 = 58.5$, Max = 61,

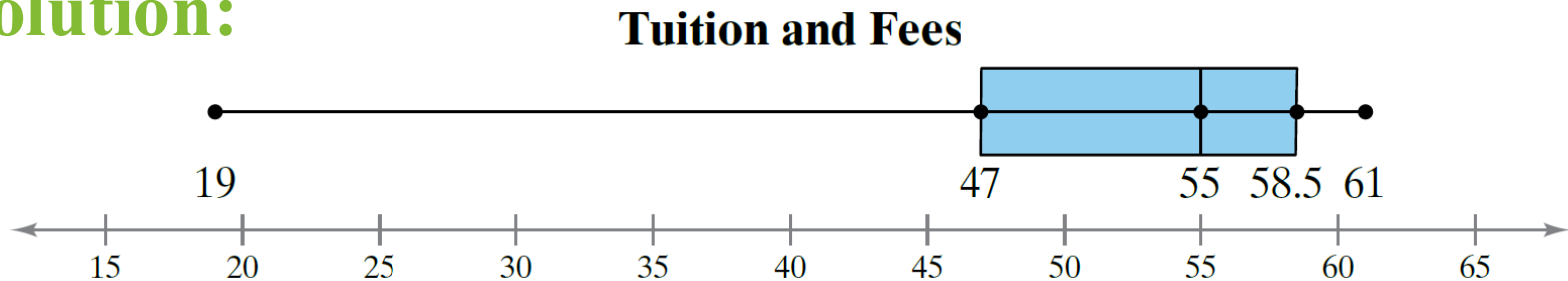
Solution:



The box represents about half of the data, which are between 47 and 58.5.

Example: Drawing a Box-and-Whisker Plot

Solution:



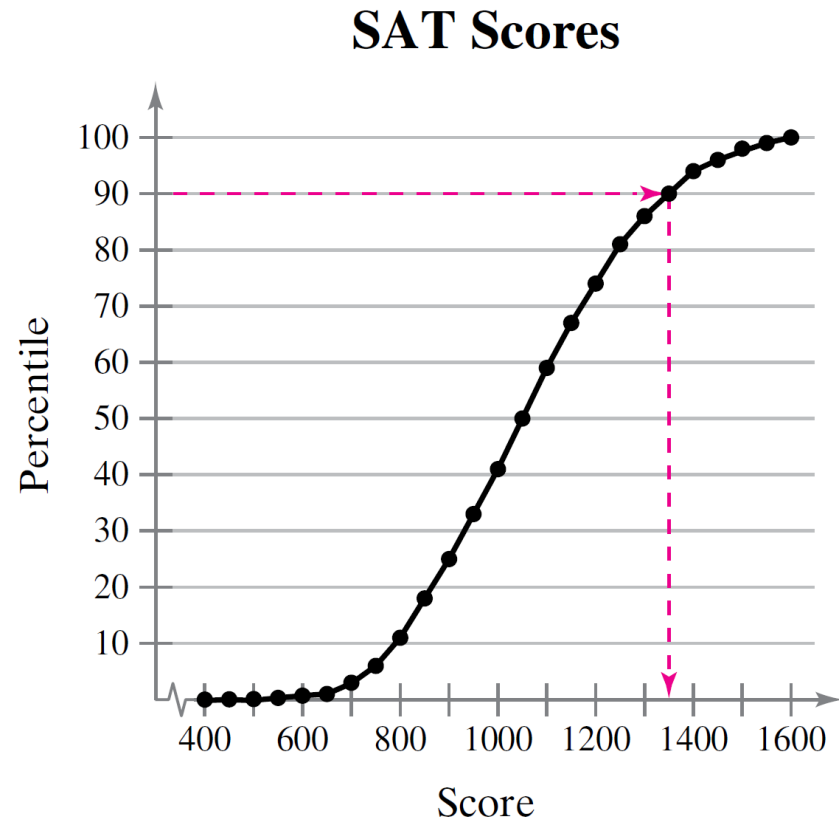
The left whisker represents about one quarter of the data, so about 25% of the data entries are less than 47. The right whisker represents about one quarter of the data, so about 25% of the data entries are greater than 58.5. Also, the length of the left whisker is much longer than the right one. This indicates that the data set has a possible outlier to the left.

Percentiles and Other Fractiles

Fractiles	Summary	Symbols
Quartiles	Divides data into 4 equal parts	Q_1, Q_2, Q_3
Deciles	Divides data into 10 equal parts	$D_1, D_2, D_3, \dots, D_9$
Percentiles	Divides data into 100 equal parts	$P_1, P_2, P_3, \dots, P_{99}$

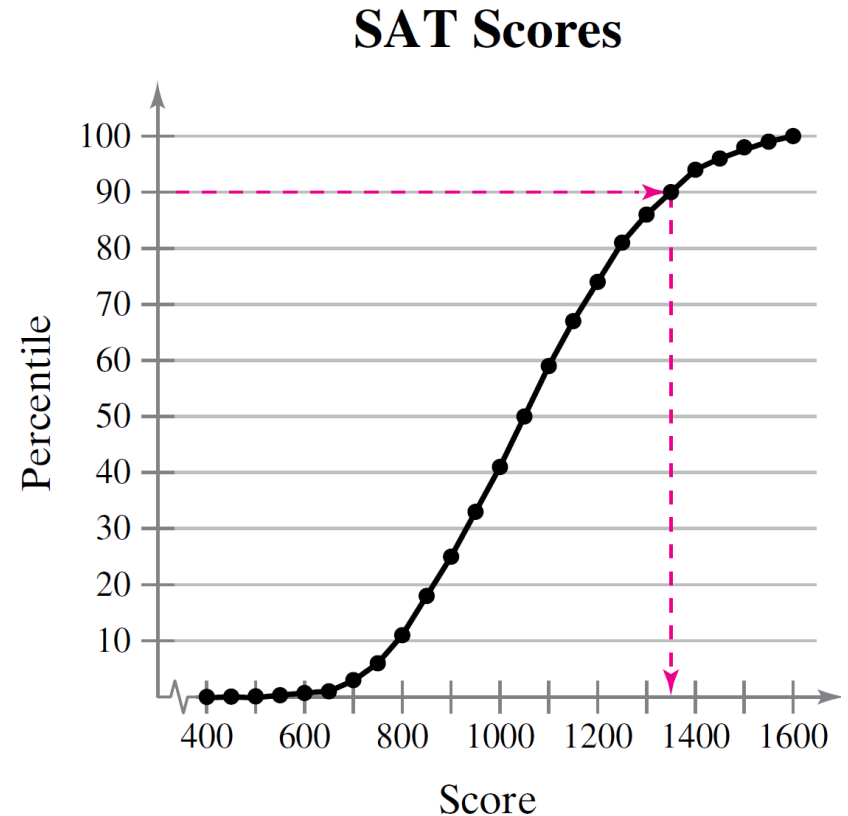
Example: Interpreting Percentiles

The ogive represents the cumulative frequency distribution for SAT test scores of college-bound students in a recent year. What test score represents the 90th percentile? (*Source: College Board*)



Solution: Interpreting Percentiles

- From the ogive, you can see that the 90th percentile corresponds to a score of 1350.
- This means that approximately 90% of the students had an SAT score of 1350 or less.



Percentile that Corresponds to a Specific Data Entry

To find the **percentile that corresponds to a specific data entry x** , use the formula

$$\text{Percentile of } x = \frac{\text{number of data entries less than } x}{\text{total number of data entries}} \cdot 100$$

and then round to the nearest whole number.

Example: Finding a Percentile

For the data set in the second example, find the percentile that corresponds to \$57,000.

Solution

- Recall that the tuition costs are in thousands of dollars, so \$57,000 is the data entry 57

19 30 39 44 45 46 48 48 48 48 55 55 55
55 56 57 57 58 58 59 59 59 60 61 61

Solution: Finding a Percentile

- There are 15 data entries less than 57 and the total number of data entries is 25.

$$\begin{aligned}\text{Percentile of } 57 &= \frac{\text{number of data entries less than } 57}{\text{total number of data entries}} \cdot 100 \\ &= \frac{15}{25} \cdot 100 = 60\end{aligned}$$

- The tuition cost of \$57,000 corresponds to the 60th percentile.

The tuition cost of \$57,000 is greater than 60% of the other tuition costs.

The Standard Score

Standard Score (z-score)

- Represents the number of standard deviations a given value x falls from the mean μ .

- $$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}} = \frac{x - \mu}{\sigma}$$

Example: Finding z-Scores

The mean speed of vehicles along a stretch of highway is 56 miles per hour with a standard deviation of 4 miles per hour. You measure the speeds of three cars traveling along this stretch of highway as 62 miles per hour, 47 miles per hour, and 56 miles per hour. Find the z-score that corresponds to each speed. Assume the distribution of the speeds is approximately bell-shaped.

Solution: Finding z-Scores

Solution

The z-score that corresponds to each speed is calculated below.

$$x = 62 \text{ mph} \quad x = 47 \text{ mph} \quad x = 56 \text{ mph}$$

$$z = \frac{62 - 56}{4} = 1.5 \quad z = \frac{47 - 56}{4} = -2.25 \quad z = \frac{56 - 56}{4} = 0$$

62 miles per hour is 1.5 standard deviations above the mean; 47 miles per hour is 2.25 standard deviations below the mean; and 56 miles per hour is equal to the mean. 47 miles per hour is unusually slow, because its speed corresponds to a z-score of -2.25 .

Example: Comparing z-Scores from Different Data Sets

The table shows the mean heights and standard deviations for a population of men and a population of women. Compare the z-scores for a 6-foot-tall man and a 6-foot-tall woman. Assume the distributions of the heights are approximately bell-shaped.

Men's heights	Women's heights
$\mu = 69.9$ in. $\sigma = 3.0$ in.	$\mu = 64.3$ in. $\sigma = 2.6$ in.

Solution: Comparing z-Scores from Different Data Sets

Solution

Note that 6 feet = 72 inches. Find the z-score for each height.

- **z-score for 6-foot-tall man**

$$z = \frac{x - \mu}{\sigma} = \frac{72 - 69.9}{3.0} = 0.7$$

- **z-score for 6-foot-tall woman**

$$z = \frac{x - \mu}{\sigma} = \frac{72 - 64.3}{2.6} \approx 3.0$$

Solution: Comparing z-Scores from Different Data Sets

Solution

The z-score for the 6-foot-tall man is within 1 standard deviation of the mean (69.9 inches). This is among the typical heights for a man.

The z-score for the 6-foot-tall woman is about 3 standard deviations from the mean (64.3 inches). This is an unusual height for a woman.

